

Detection recall vs. coverage

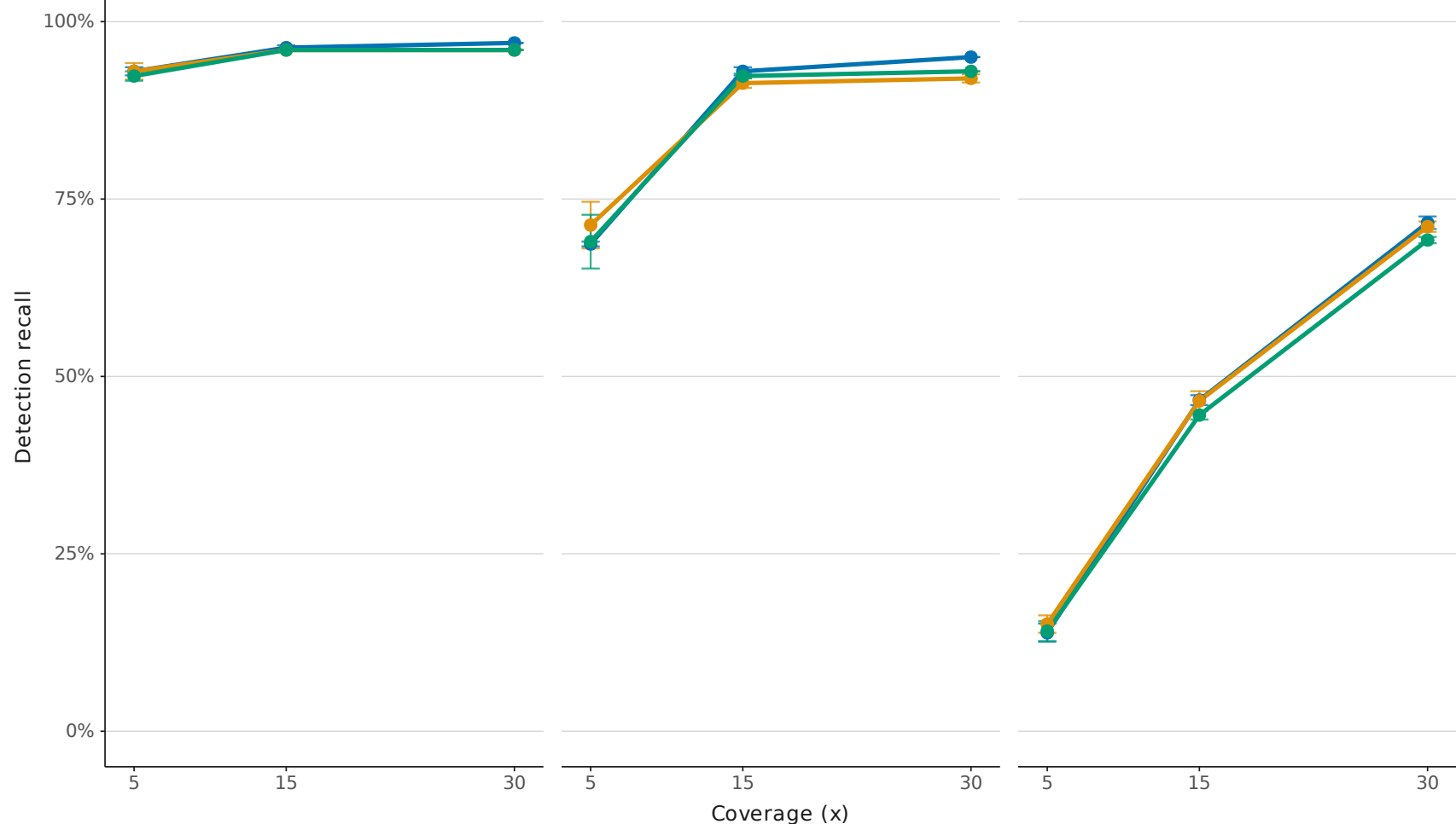
Fraction of simulated insertions recovered (mean $\pm$ SEM, 3 replicates)

● RelocaTE2 ● RelocaTE3-bwa/bwa ● RelocaTE3-minimap2/minimap2

Homozygous

Heterozygous

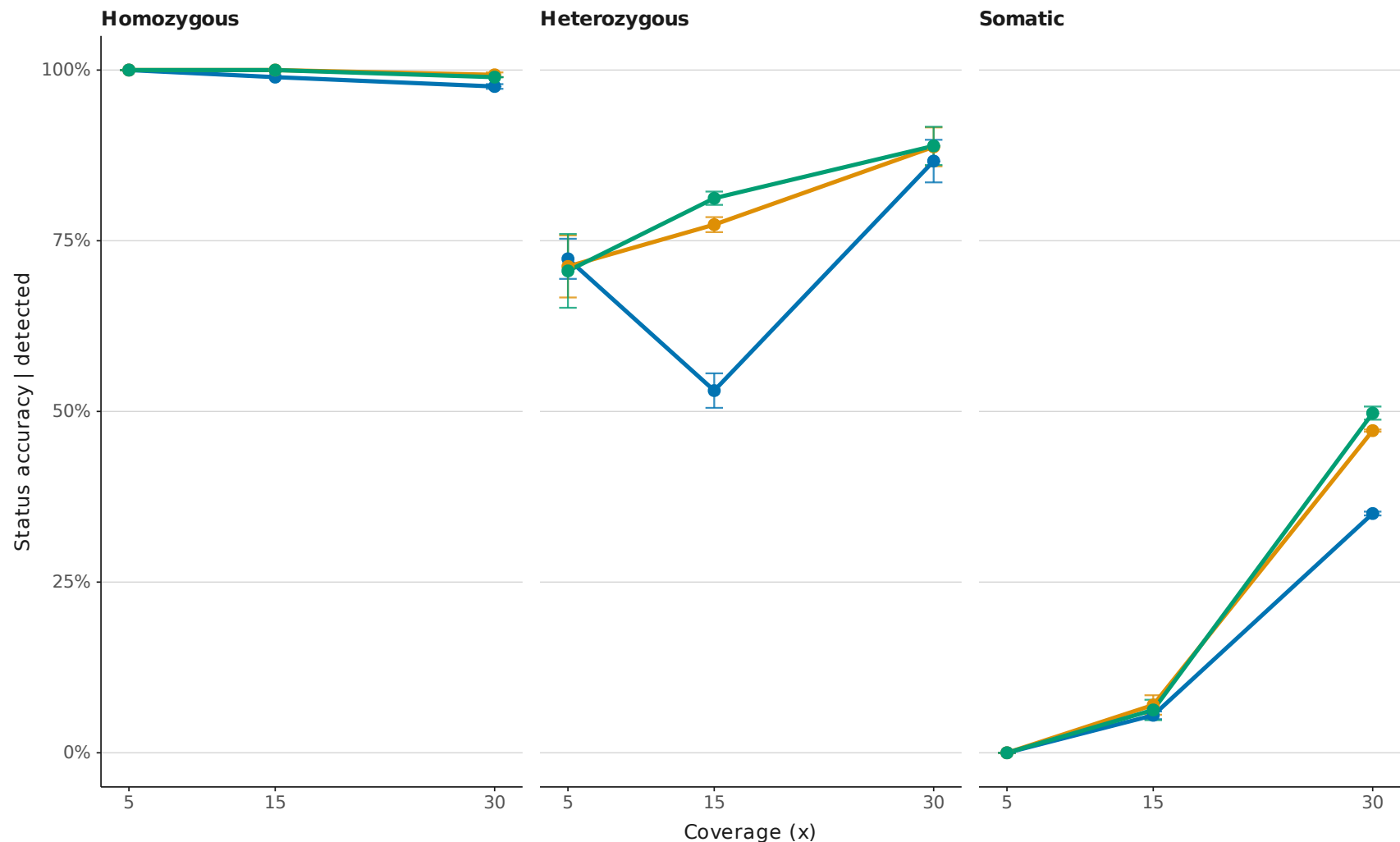
Somatic



Genotype-status accuracy vs. coverage

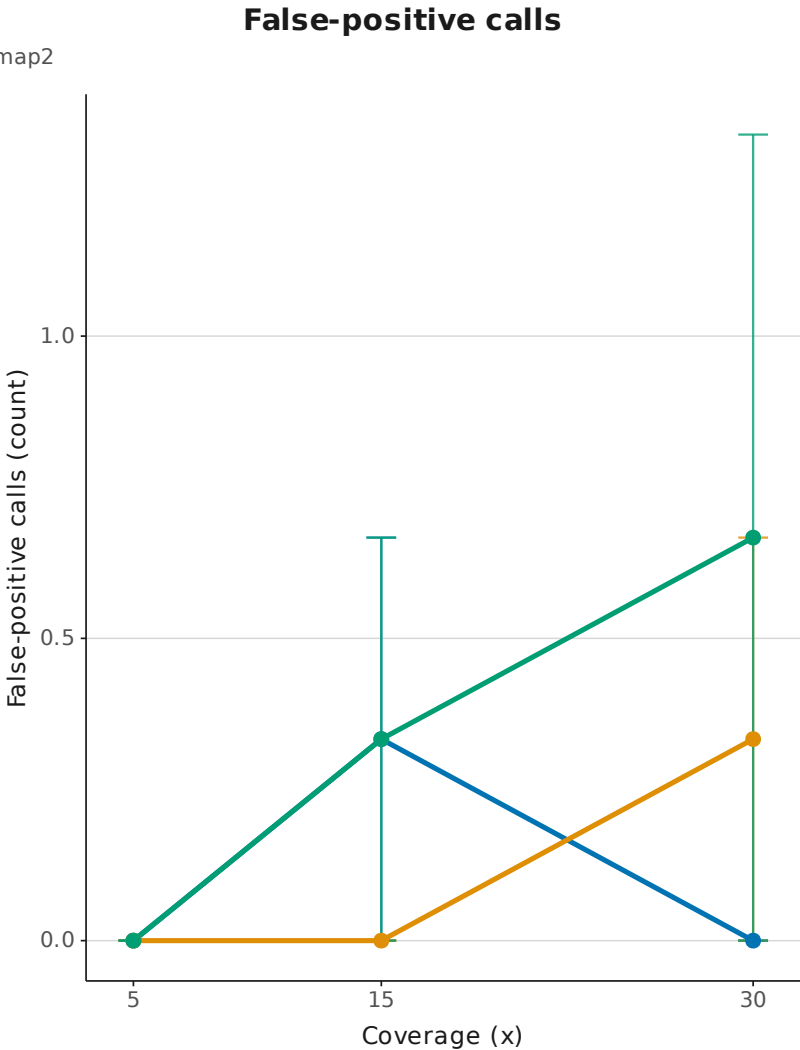
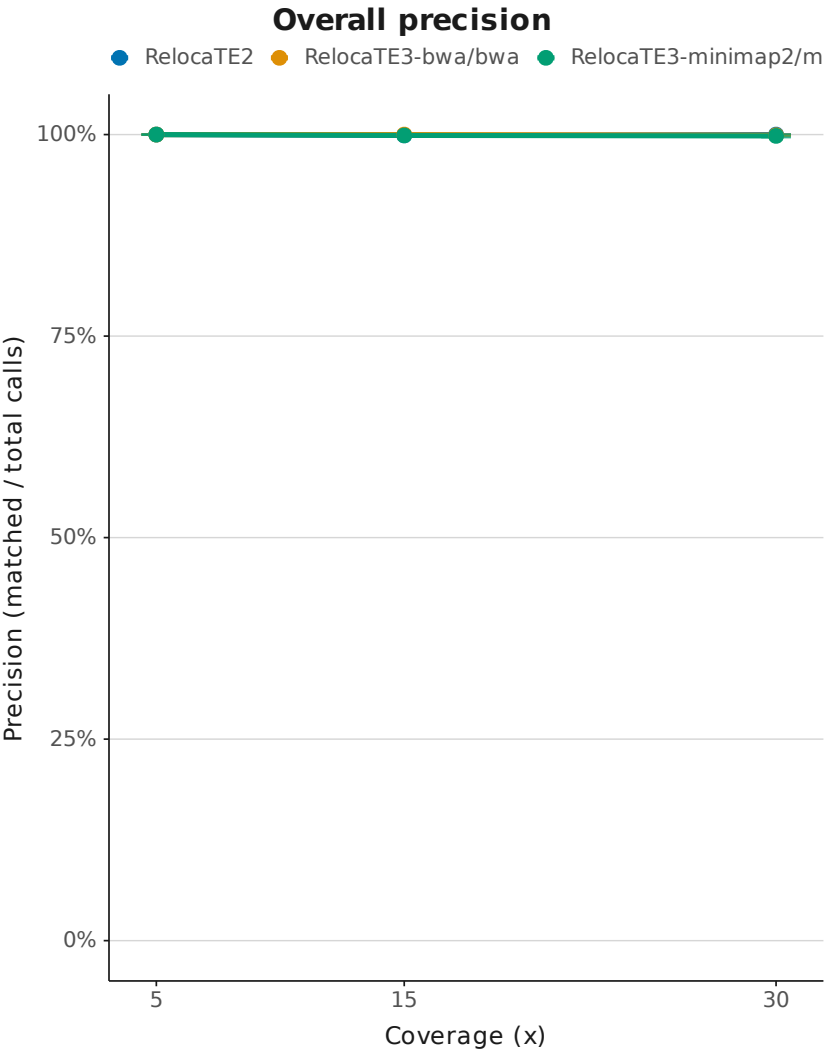
Correct hom/het/somatic call among DETECTED insertions (mean $\pm$ SEM)

RelocaTE2 RelocaTE3-bwa/bwa RelocaTE3-minimap2/minimap2



Precision and false positives vs. coverage

RelocaTE precision = fraction of calls matching a true insertion (mean $\pm$ SEM)



Somatic detection recall by cellular fraction

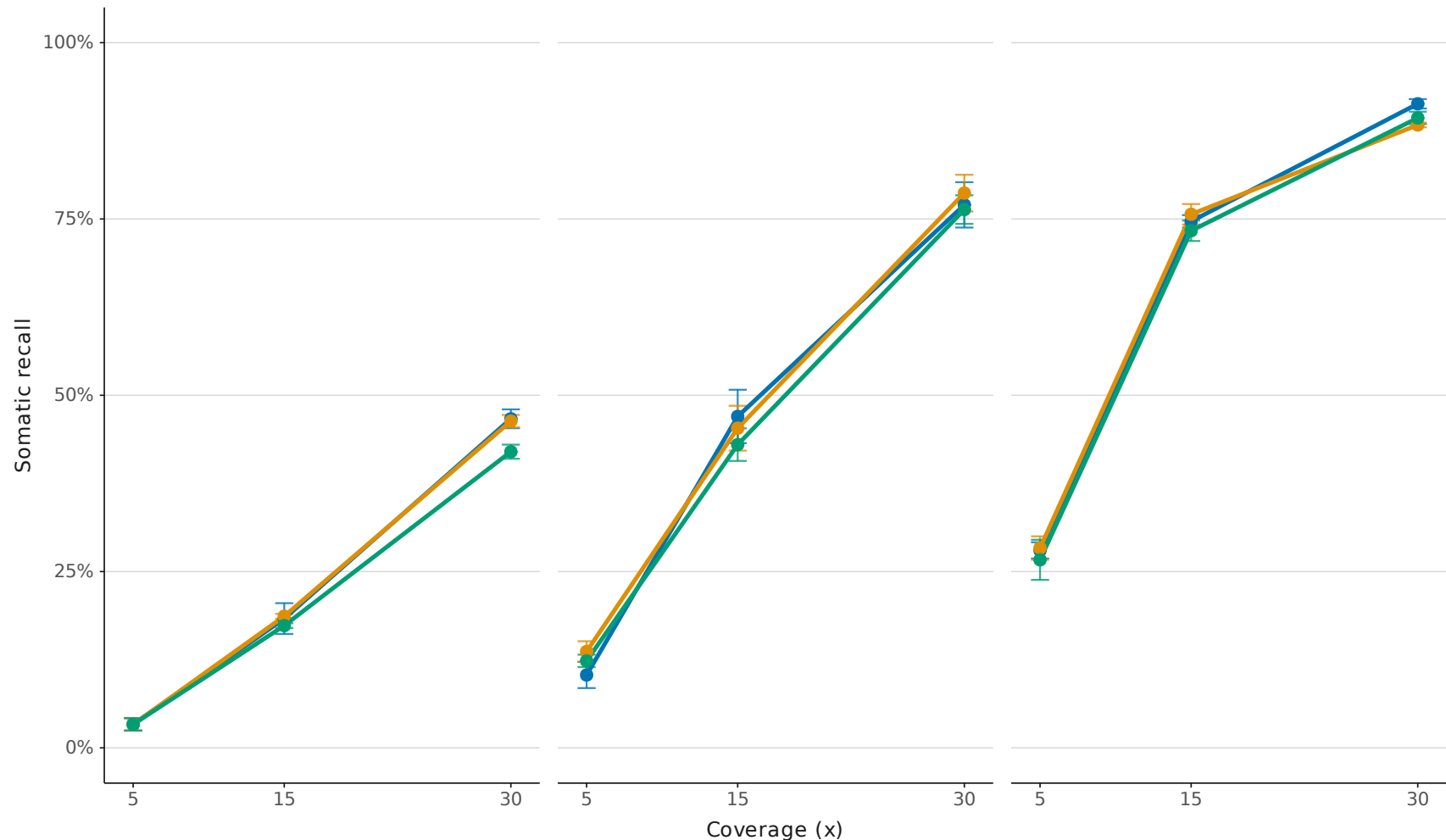
Lower cellular fraction = lower variant allele frequency = harder (mean $\pm$ SEM)

● RelocaTE2 ● RelocaTE3-bwa/bwa ● RelocaTE3-minimap2/minimap2

Cellular fraction 0.1 (VAF 0.05)

Cellular fraction 0.2 (VAF 0.1)

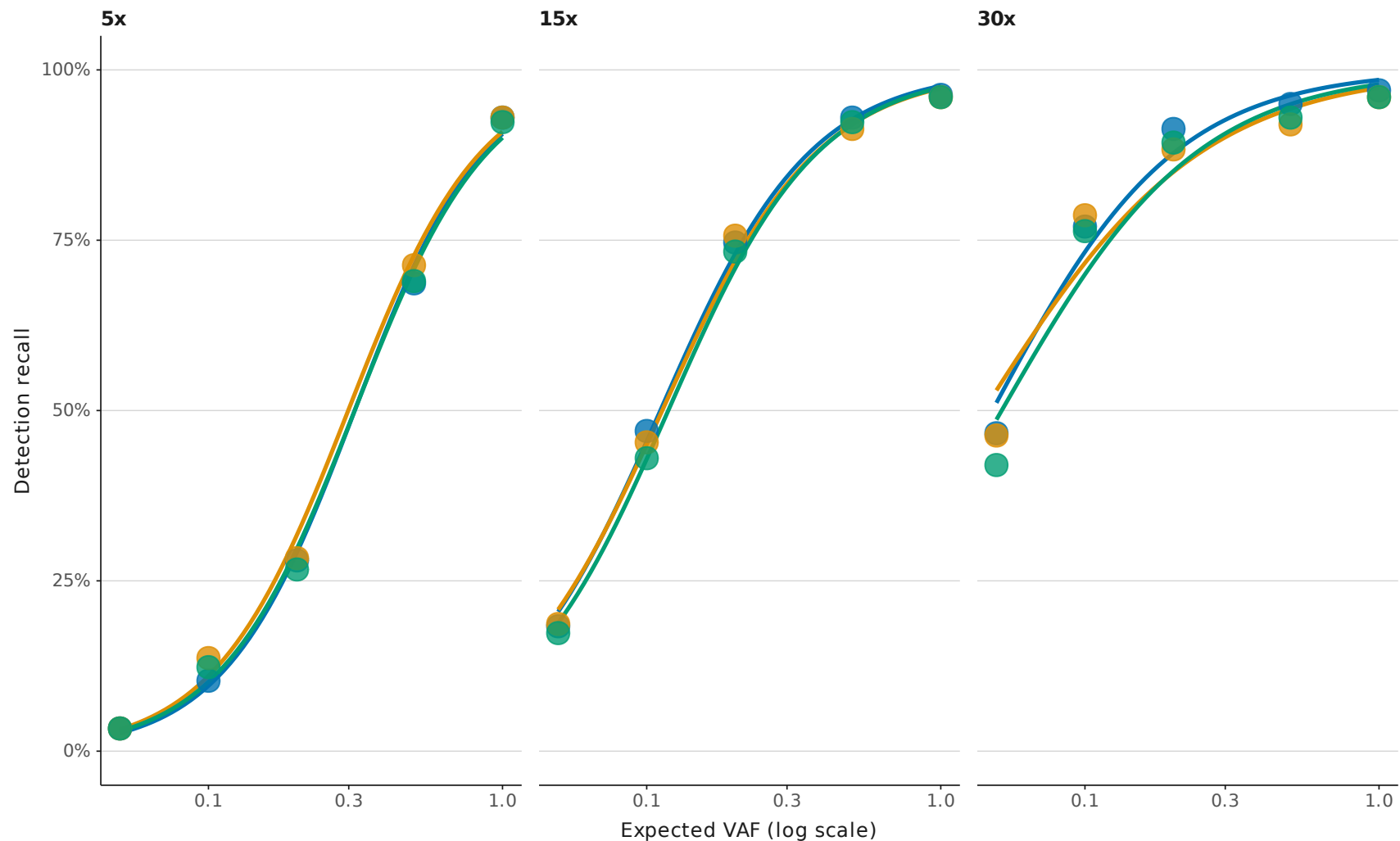
Cellular fraction 0.4 (VAF 0.2)



Detection sensitivity vs variant allele frequency

Logistic fit per caller; VAF at 50% detection = limit of detection (LOD50)

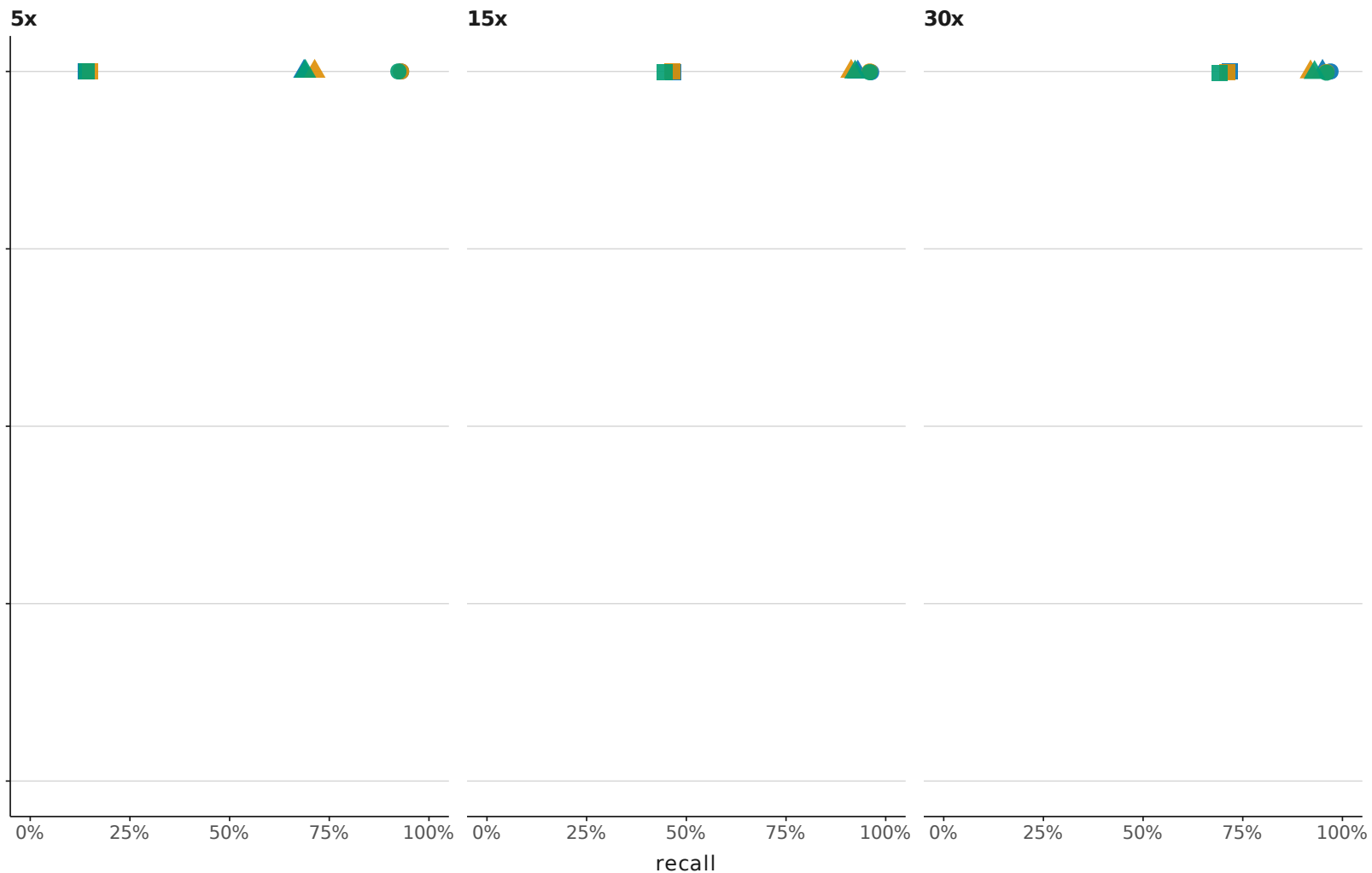
RelocaTE2 RelocaTE3-bwa/bwa RelocaTE3-minimap2/minimap2



Precision-recall operating points

Precision = matched / all calls per sample (global denominator); one point per class

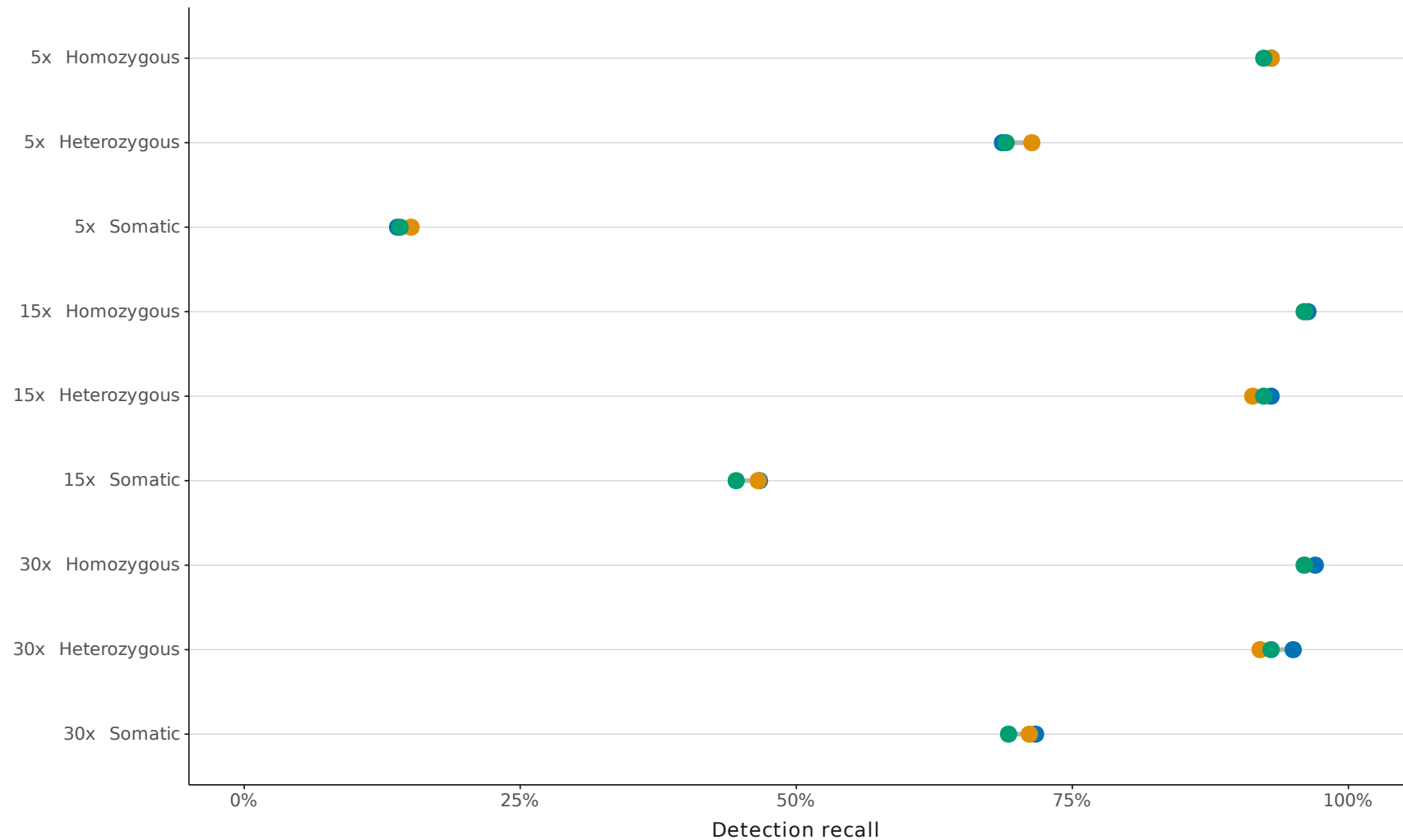
● Homozygous ▲ Heterozygous ■ Somatic ● RelocaTE2 ● RelocaTE3-bwa/bwa ● RelocaTE3-minimap2/minimap2



Detection recall by caller

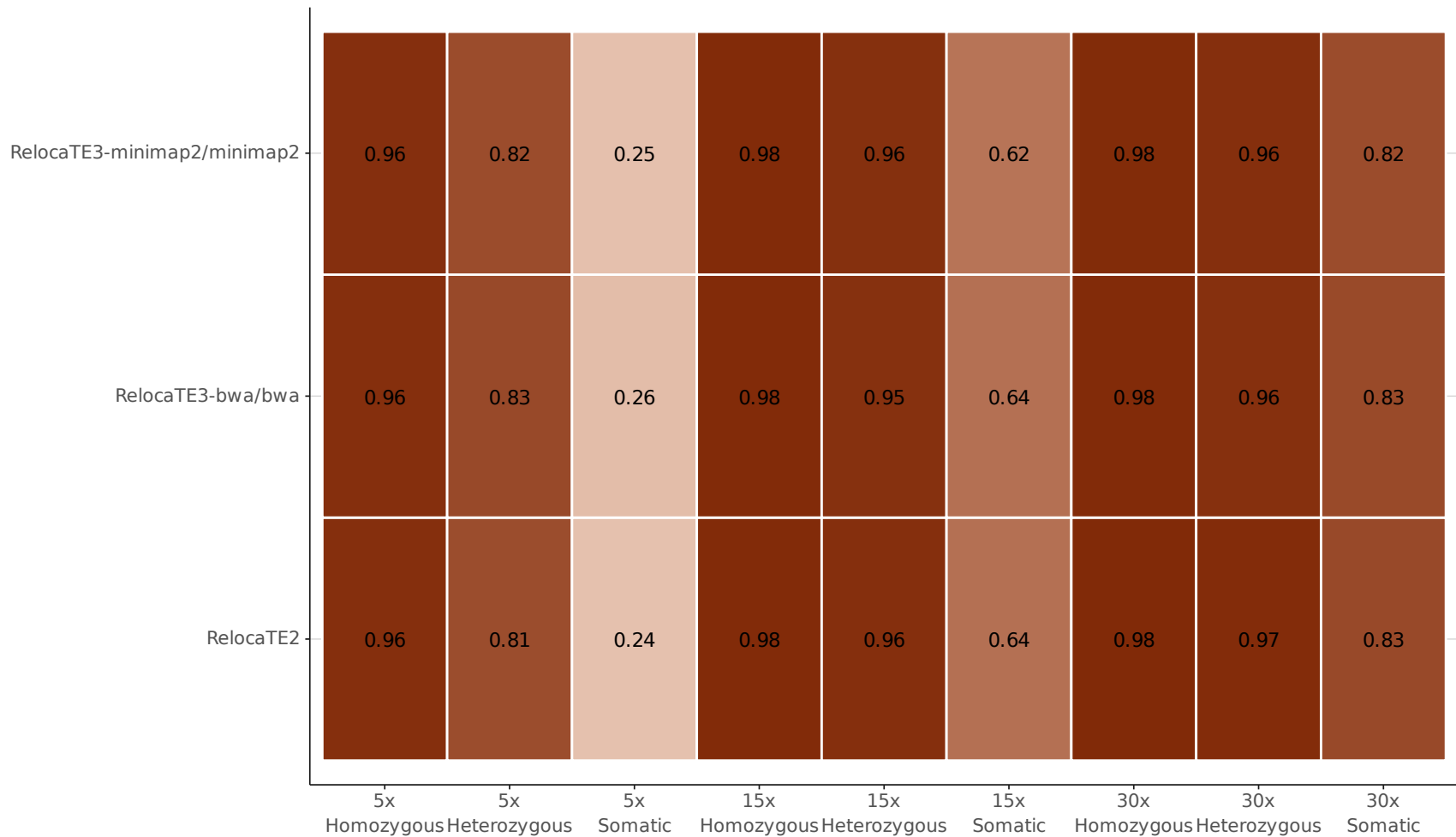
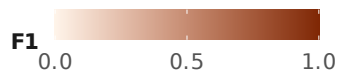
Each row = coverage x class; grey connector spans the 3 callers

● RelocaTE2 ● RelocaTE3-bwa/bwa ● RelocaTE3-minimap2/minimap2



F1 (detection x precision) by condition

Harmonic mean of recall and precision (precision = global-denominator, per sample)



Genotype / detection confusion

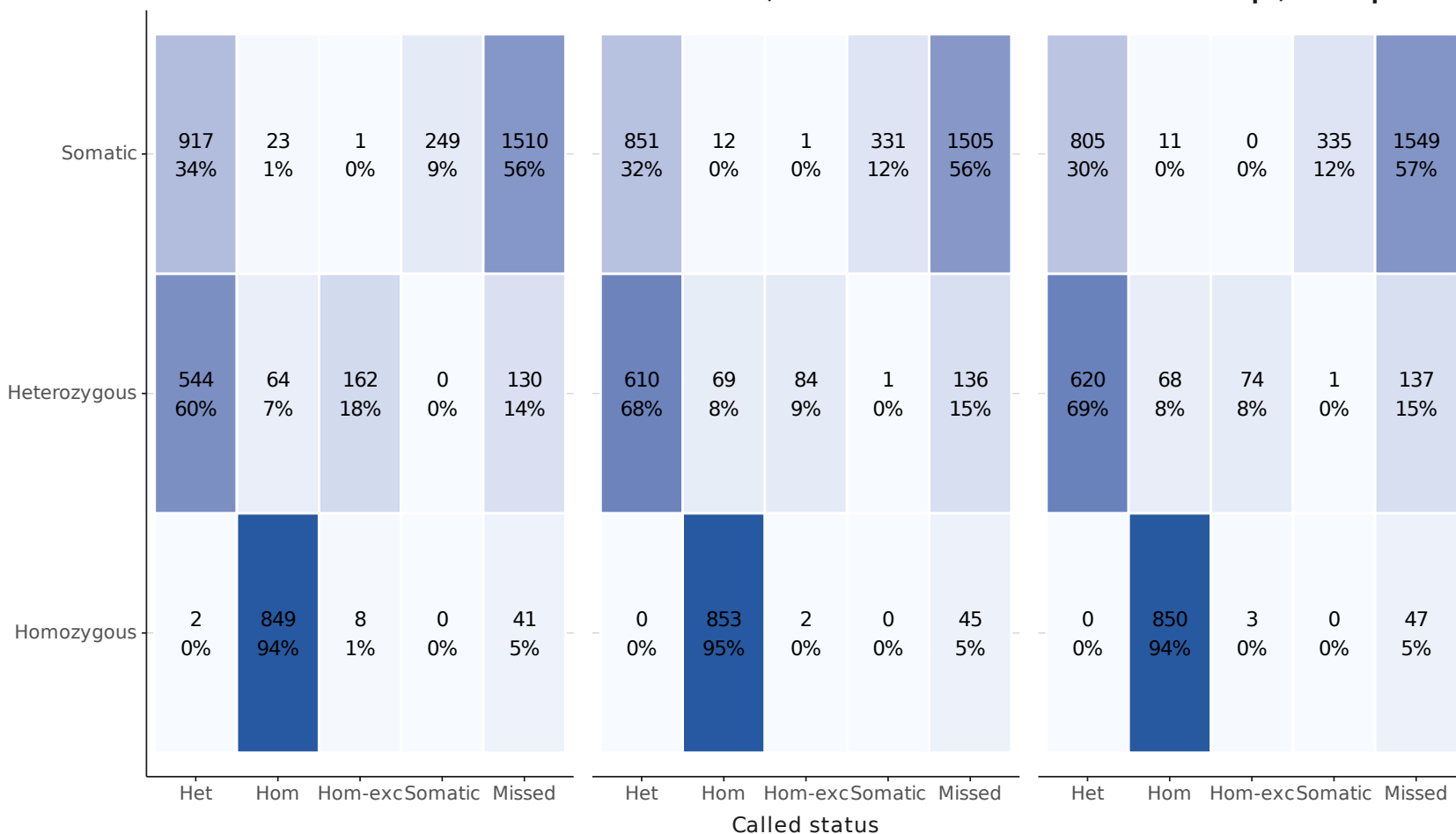
Row-normalised: where each truth class's events land (pooled over coverage)

Row %
0% 50% 100%

RelocaTE2

RelocaTE3-bwa/bwa

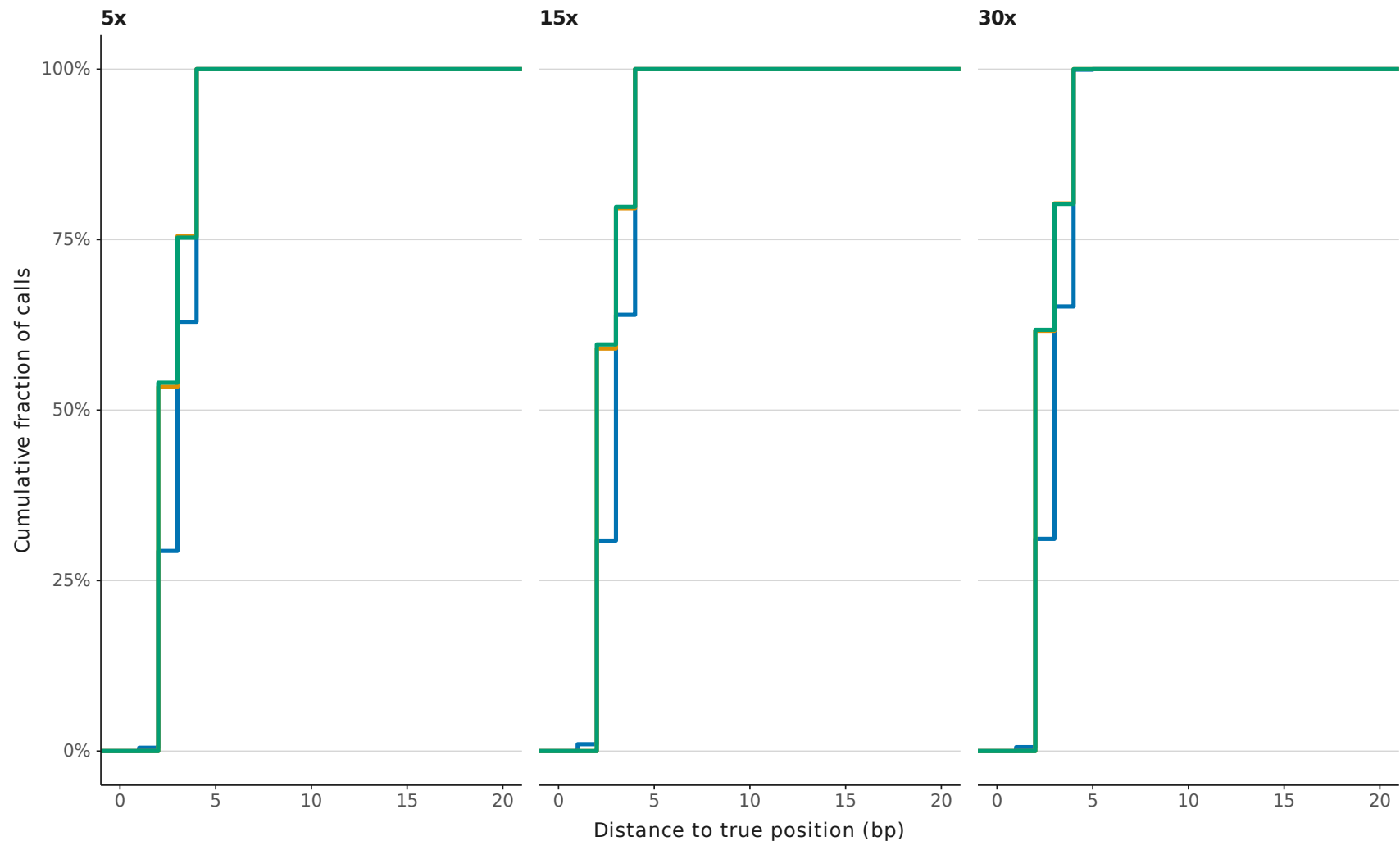
RelocaTE3-minimap2/minimap2



Breakpoint positional accuracy

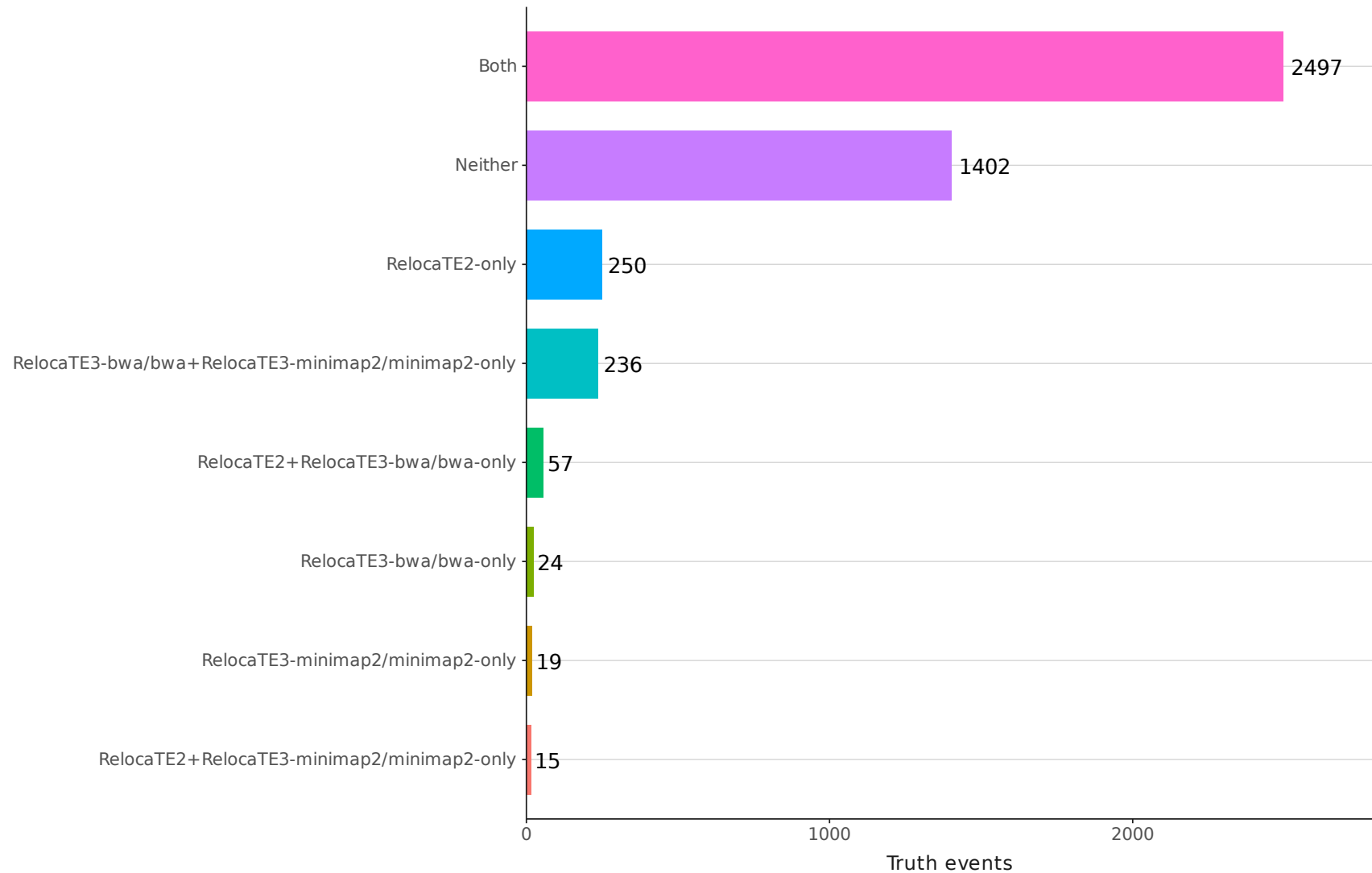
ECDF of $|\text{called} - \text{true}|$ position for matched events; steeper/left = better

RelocaTE2 RelocaTE3-bwa/bwa RelocaTE3-minimap2/minimap2



Per-event caller agreement

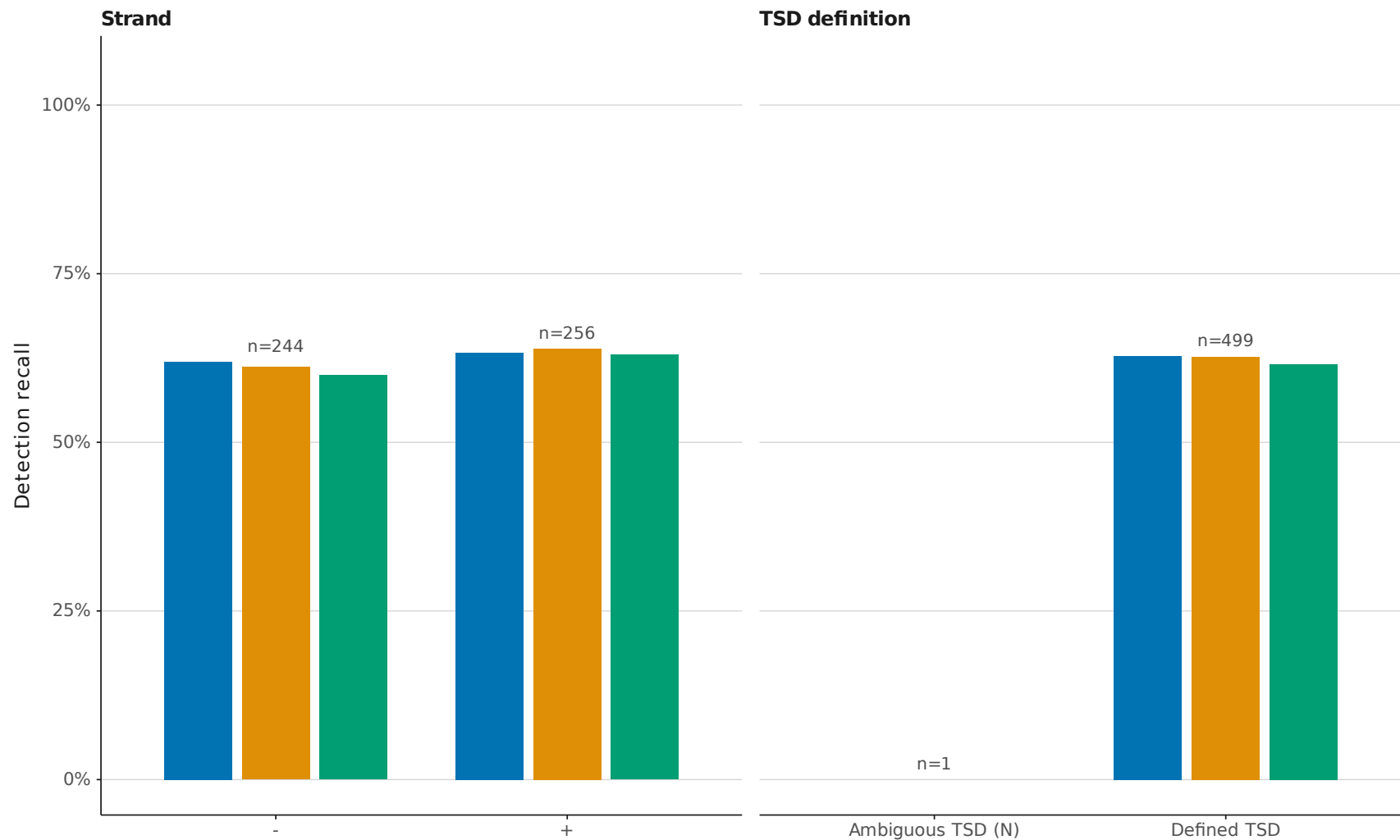
Truth events by which caller(s) detected them (pooled over all samples)



What do missed events have in common?

Detection recall stratified by strand and TSD ambiguity (pooled); n = distinct truth events

RelocaTE2 RelocaTE3-bwa/bwa RelocaTE3-minimap2/minimap2



Compute resources vs. coverage

Per-sample wall time and peak RSS (mean $\pm$ SEM)

